



File Processing



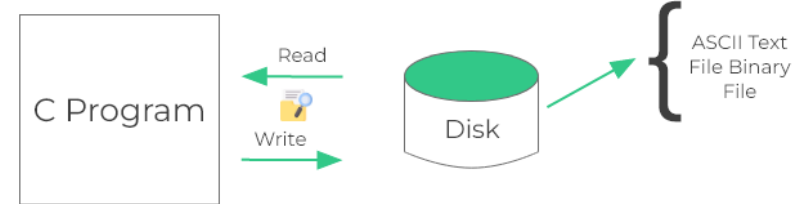
File Introduction

- Files are places where data can be stored permanently.
 - Some programs expect the same set of data to be fed as input every time it is run.
 - Cumbersome.
 - Better if the data are kept in a file, and the program reads from the file.
-

Text File vs Binary File

File Handling in PASCAL

Based on “ASCII code”



Text Files

- When using a **text** file, we write out separately each of the pieces of data about a given record.
- The text file will be **readable** by an editor

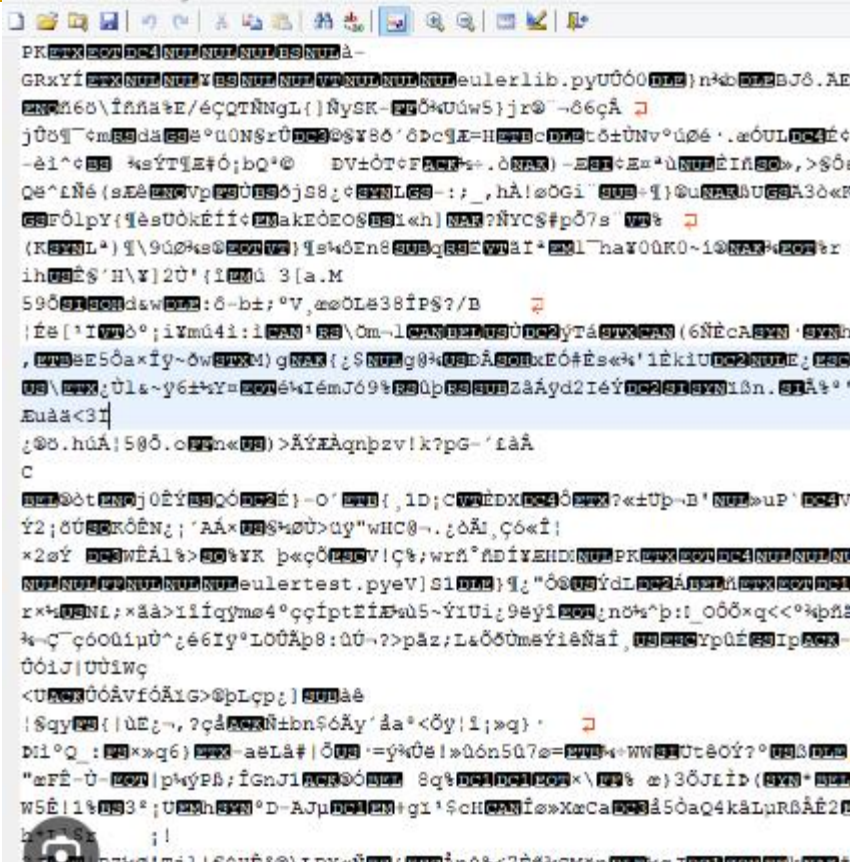
Binary Files

- When using a **binary** file we write whole record data to the file at once.
- but the numbers in the binary file will **not** be **readable** in this way.

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		DEL

[illegible]

249	•
250	•
251	✓
252	n
253	2
254	■
255	



```
line-1
line-2
line-3
line-4
```

Text File
readable

Binary File
unreadable

File Edit Options Encoding Help

line-1
line-2
line-3
line-4

--- SAMPLE 1

#include <iostream>

int main()

```
{
char ch;
FILE *fp1 = fopen( "D:/mydoc/cos2101/DemoCOS2101/Debug/test1.txt", "rt");
if (fp1 == NULL) {
printf("cannot open file !!");
return 0;
}
do {
long pos = ftell(fp1);
ch = getc(fp1);
printf("%c \t %x \t %d \n",ch, ch , pos);
} while (ch != EOF);
```

fclose(fp1);

}

TEXT FILE

ปกติจะมีรหัส 0xD, 0xA

เพื่อเป็นตัวควบคุมการเว้นบรรทัด

File Edit Options Encoding Help

```
00000000: 6C 69 6E 65 2D 31 0D 0A|6C 69 6E 65 2D 32 0D 0A |
00000010: 6C 69 6E 65 2D 33 0D 0A|6C 69 6E 65 2D 34 0D 0A |
00000020: 0D 0A 2D 2D 2D 20 53 41|4D 50 4C 45 20 31 0D 0A |
00000030: 23 69 6E 63 6C 75 64 65|20 3C 69 6F 73 74 72 65 |
00000040: 61 6D 3E 0D 0A 0D 0A 69|6E 74 20 6D 61 69 6E 28 |
00000050: 29 0D 0A 7B 0D 0A 63 68|61 72 20 63 68 3B 0D 0A |
00000060: 46 49 4C 45 20 2A 66 70|31 20 3D 20 66 6F 70 65 |
00000070: 6E 28 20 22 44 3A 2F 6D|79 64 6F 63 2F 63 6F 73 |
00000080: 32 31 30 31 2F 44 65 6D|6F 43 4F 53 32 31 30 31 |
00000090: 2F 44 65 62 75 67 2F 74|65 73 74 31 2E 74 78 74 |
000000A0: 22 2C 20 22 72 74 22 29|3B 0D 0A 69 66 20 28 66 |
000000B0: 70 31 20 3D 3D 20 4E 55|4C 4C 29 20 7B 0D 0A 70 |
000000C0: 72 69 6E 74 66 28 22 63|61 6E 6E 6F 74 20 6F 70 |
000000D0: 65 6E 20 66 69 6C 65 20|21 21 22 29 3B 0D 0A 72 |
000000E0: 65 74 75 72 6E 20 30 3B|0D 0A 7D 0D 0A 64 6F 20 |
000000F0: 7B 0D 0A 6C 6F 6E 67 20|70 6F 73 20 3D 20 66 74 |
00000100: 65 6C 6C 28 66 70 31 29|3B 0D 0A 63 68 20 3D 20 |
00000110: 67 65 74 63 28 66 70 31|29 3B 0D 0A 70 72 69 6E |
00000120: 74 66 28 22 25 63 20 5C|74 20 20 25 78 20 5C 74 |
00000130: 20 25 64 20 5C 6E 22 2C|63 68 2C 20 63 68 20 2C |
00000140: 20 70 6F 73 29 3B 0D 0A|7D 20 77 68 69 6C 65 20 |
00000150: 28 63 68 20 21 3D 20 45|4F 46 29 3B 0D 0A 0D 0A |
00000160: 66 63 6C 6F 73 65 28 66|70 31 29 3B 0D 0A 0D 0A |
00000170: 7D 0D 0A 0D 0A 2D 2D 20|53 41 4D 50 4C 45 20 32 |
00000180: 0D 0A 23 64 65 66 69 6E|65 20 5F 43 52 54 5F 53 |
00000190: 45 43 55 52 45 5F 4E 4F|5F 57 41 52 4E 49 4E 47 |
000001A0: 53 0D 0A 23 69 6E 63 6C|75 64 65 20 3C 69 6F 73 |
000001B0: 74 72 65 61 6D 3E 0D 0A|0D 0A 0D 0A 69 6E 74 20 |
000001C0: 6D 61 69 6E 28 29 0D 0A|7B 0D 0A 63 68 61 72 20 |
000001D0: 63 68 3B 0D 0A 46 49 4C|45 20 2A 66 70 31 20 3D |
000001E0: 20 66 6F 70 65 6E 28 22|44 3A 2F 6D 79 64 6F 63 |
000001F0: 2F 63 6F 73 32 31 30 31|2F 44 65 6D 6F 43 4F 53 |
00000200: 32 31 30 31 2F 44 65 62|75 67 2F 74 65 73 74 31 |
00000210: 2E 74 78 74 22 2C 20 22|72 74 22 29 3B 0D 0A 69 |
00000220: 66 20 28 66 70 31 20 3D|3D 20 4E 55 4C 4C 29 20 |
00000230: 7B 0D 0A 70 72 69 6E 74|66 28 22 63 61 6E 6E 6F |
00000240: 74 20 6F 70 65 6E 20 66|69 6C 65 20 21 21 22 29 |
00000250: 3B 0D 0A 72 65 74 75 72|6E 20 30 3B 0D 0A 7D 0D |
00000260: 0A 0D 0A 64 6F 20 7B 0D|0A 6C 6F 6E 67 20 70 6F |
```

```
| line-1..line-2..
| line-3..line-4..
| ---- SAMPLE 1..
| #include <iostre
| am>....int main(
| )..{..char ch;..
| FILE *fp1 = fope
| n( "D:/mydoc/cos
| 2101/DemoCOS2101
| /Debug/test1.txt
| ", "rt");...if (f
| p1 == NULL) {...p
| rintf("cannot op
| en file !!");..r
| eturn 0;...}..do
| {...long pos = ft
| ell(fp1);..ch =
| getc(fp1);..prin
| tf("%c \t %x \t
| %d \n",ch, ch ,
| pos);..} while
| (ch != EOF);....
| fclose(fp1);....
| }....-- SAMPLE 2
| ..#define _CRT_S
| ECURE_NO_WARNING
| S..#include <ios
| tream>.....int
| main()..{..char
| ch;..FILE *fp1 =
| fopen("D:/mydoc
| /cos2101/DemoCOS
| 2101/Debug/test1
| .txt", "rt");..i
| f (fp1 == NULL)
| {...printf("canno
| t open file !!")
| ;..return 0;...}
| ...do {...long po
```

File	Edit	Options	Encoding	Help
%PDF-1.5	00000000: 25 50 44 46 20 31 2E 35 0D 0A 25 B5 B5 B5 B5 0D	%PDF-1.5..%nnnn.		
%nnnn	00000010: 0A 31 20 30 20 6F 62 6A 0D 0A 3C 3C 2F 54 79 70	.1 0 obj.<</Typ		
1 0 obj	00000020: 65 2F 43 61 74 61 6C 6F 67 2F 50 61 67 65 73 20	e/Catalog/Pages		
<</Type/Catalog/Pages 2 0 R/Lang(en-US) /StructTree	00000030: 32 20 30 20 52 2F 4C 61 6E 67 28 65 6E 2D 55 53	2 0 R/Lang(en-US		
true>>/Metadata 1478 0 R/ViewerPreferences 1479 0	00000040: 29 20 2F 53 74 72 75 63 74 54 72 65 65 52 6F 6F	) /StructTreeRoo		
endobj	00000050: 74 20 36 34 20 30 20 52 2F 4D 61 72 6B 49 6E 66	t 64 0 R/MarkInf		
2 0 obj	00000060: 6F 3C 3C 2F 4D 61 72 6B 65 64 20 74 72 75 65 3E	o<</Marked true>		
<</Type/Pages/Count 11/Kids[3 0 R 20 0 R 38 0 R 40	00000070: 3E 2F 4D 65 74 61 64 61 74 61 20 31 34 37 38 20	>>/Metadata 1478		
R 58 0 R 60 0 R 62 0 R] >>	00000080: 30 20 52 2F 56 69 65 77 65 72 50 72 65 66 65 72	0 R/ViewerPrefer		
endobj	00000090: 65 6E 63 65 73 20 31 34 37 39 20 30 20 52 3E 3E	ences 1479 0 R>>		
3 0 obj	000000A0: 0D 0A 65 6E 64 6F 62 6A 0D 0A 32 20 30 20 6F 62	..endobj..2 0 ob		
<</Type/Page/Parent 2 0 R/Resources<</ExtGState<</	000000B0: 6A 0D 0A 3C 3C 2F 54 79 70 65 2F 50 61 67 65 73	j..<</Type/Pages		
R>>/Font<</F1 6 0 R/F2 12 0 R/F3 14 0 R/F4 16 0 R/	000000C0: 2F 43 6F 75 6E 74 20 31 31 2F 4B 69 64 73 5B 20	/Count 11/Kids[		
R>>/ProcSet[/PDF/Text/ImageB/ImageC/ImageI] >>/Media	000000D0: 33 20 30 20 52 20 32 30 20 30 20 52 20 33 38 20	3 0 R 20 0 R 38		
0 R/Group<</Type/Group/S/Transparency/CS/DeviceRGB	000000E0: 30 20 52 20 34 30 20 30 20 52 20 34 34 20 30 20	0 R 40 0 R 44 0		
endobj	000000F0: 52 20 35 31 20 30 20 52 20 35 34 20 30 20 52 20	R 51 0 R 54 0 R		
4 0 obj	00000100: 35 36 20 30 20 52 20 35 38 20 30 20 52 20 36 30	56 0 R 58 0 R 60		
<</Filter/FlateDecode/Length 889>>	00000110: 20 30 20 52 20 36 32 20 30 20 52 5D 20 3E 3E 0D	0 R 62 0 R] >>.		
stream	00000120: 0A 65 6E 64 6F 62 6A 0D 0A 33 20 30 20 6F 62 6A	.endobj..3 0 obj		
xaXKo 0w00n-n-0-[@` M1=3 `0`a`mb`b 1JN4\$u*S`0njQ`d0	00000130: 0D 0A 3C 3C 2F 54 79 70 65 2F 50 61 67 65 2F 50	..<</Type/Page/P		
61w]?	00000140: 61 72 65 6E 74 20 32 20 30 20 52 2F 52 65 73 6F	arent 2 0 R/Reso		
Ai1UE-`w9p^0x5x+au ( 100uu{UWb~90b] 4q~d<?_0`?0G	00000150: 75 72 63 65 73 3C 2F 45 78 74 47 53 74 61 74	urces<</ExtGStat		
0-0.00003q_W00n\$<wZ,n`mno0'5n0X ~-Jw`xSW_0m0\$000i	00000160: 65 3C 3C 2F 47 53 35 20 35 20 30 20 52 2F 47 53	e<</GS5 5 0 R/GS		
j.i`m0a1w1 Ba0` m00\$0m00-0m0Wk+<ZXfu-}aT`00u0D>~0o x0	00000170: 31 31 20 31 31 20 30 20 52 3E 3E 2F 46 6F 6E 74	11 11 0 R>>/Font		
0j000\$0000 0#00000` 000000` !*0000+000000	00000180: 3C 3C 2F 46 31 20 36 20 30 20 52 2F 46 32 20 31	<</F1 6 0 R/F2 1		
00`",n?,Y0n00eMm 0H00~a0-M U1)0[` 00{ jS?WP?0S000000	00000190: 32 20 30 20 52 2F 46 33 20 31 34 20 30 20 52 2F	2 0 R/F3 14 0 R/		
0-00 0a90:h=00T00S00sp`0Ca0 0\0(00~ (tA00000.0\$000000	000001A0: 46 34 20 31 36 20 30 20 52 2F 46 35 20 31 38 20	F4 16 0 R/F5 18		
0de-000(u06 0F0L Jj 00(0J000,0m'0p!00000_400000000	000001B0: 30 20 52 3E 3E 2F 50 72 6F 63 53 65 74 5B 2F 50	0 R>>/ProcSet[/P		
0F\$000J R000u\00 R`Q00000^T0000000B0W0000000000000	000001C0: 44 46 2F 54 65 78 74 2F 49 6D 61 67 65 42 2F 49	DF/Text/ImageB/I		
0ri,000-00000000000 \$000A0000000000000000000000000	000001D0: 6D 61 67 65 43 2F 49 6D 61 67 65 49 5D 20 3E 3E	mageC/ImageI] >>		
00	000001E0: 2F 4D 65 64 69 61 42 6F 78 5B 20 30 20 30 20 36	/MediaBox[0 0 6		
G_00	000001F0: 31 32 20 37 39 32 5D 20 2F 43 6F 6E 74 65 6E 74	12 792] /Content		
c`B0` ~00	00000200: 73 20 34 20 30 20 52 2F 47 72 6F 75 70 3C 3C 2F	s 4 0 R/Group<</		
c000x83000000,~0 2b-0000000000000000000000000000000	00000210: 54 79 70 65 2F 47 72 6F 75 70 2F 53 2F 54 72 61	Type/Group/S/Tra		
- 0,0/00/0I00	00000220: 6E 73 70 61 72 65 6E 63 79 2F 43 53 2F 44 65 76	nsparency/CS/Dev		
endstream	00000230: 69 63 65 52 47 42 3E 3E 2F 54 61 62 73 2F 53 2F	iceRGB>>/Tabs/S/		
endobj	00000240: 53 74 72 75 63 74 50 61 72 65 6E 74 73 20 30 3E	StructParents 0>		
5 0 obj	00000250: 3E 0D 0A 65 6E 64 6F 62 6A 0D 0A 34 20 30 20 6F	>..endobj..4 0 o		
<</Type/ExtGState/BM/Normal/ca 1>>				
endobj				
6 0 obj				
<</Type/Font/Subtype/Type0/BaseFont/BCDEEE+THSarabunPSK-Bold/Encoding/Identity-H/				
DescendantFonts 7 0 R/ToUnicode 1454 0 R>>				
endobj				

การอ่านข้อมูลจาก array of character

■ Array of character

Type

Index: Integer

c1,c2,c3:character

str : string

str := "HOME";

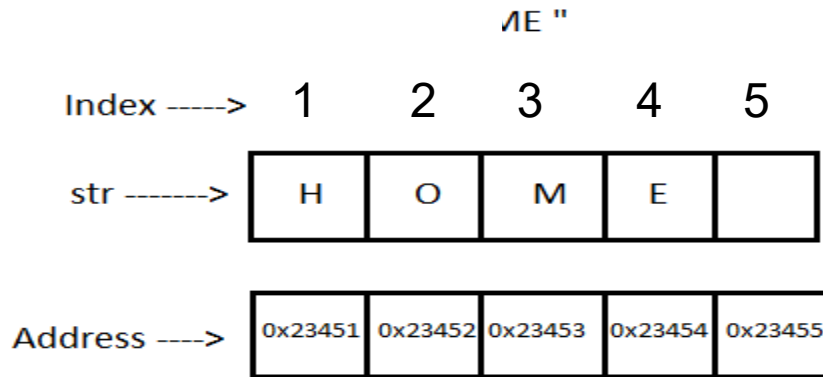
Index := 1;

c1 := str[index];
index := index+1;

c2 = str[index];
index := index+1;

Index := 4;

c3 = str[index]
index := index+1;

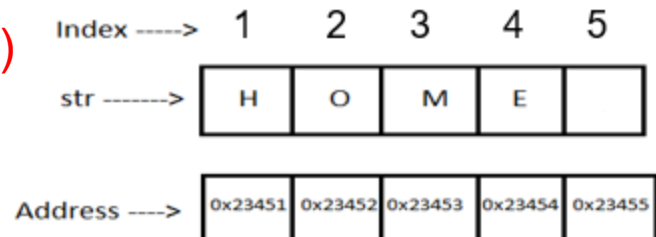


index identifies location of elements on array to do such as read and write, called access.

Read 'H'
adjust position forward (2)

Read 'O'
adjust position forward (3)

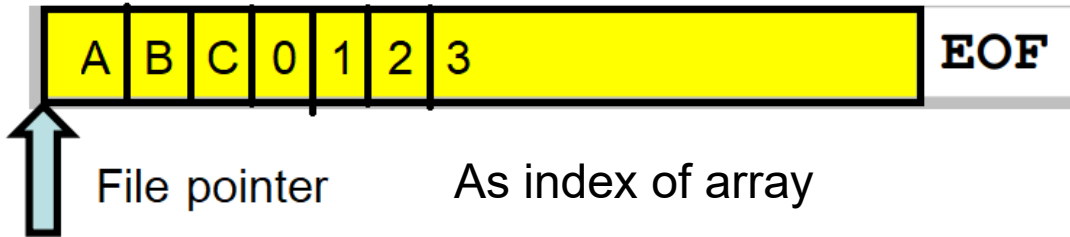
Write 'E'
adjust position backward



File Concepts

1. มอง file เป็น array of character
2. มี File pointer /Find Handle ทำหน้าที่เป็นตัวชี้ตำแหน่งข้อมูลปัจจุบัน
3. ทุกครั้งที่มีการอ่าน(read/write) file pointer จะขยับมา 1 ตัวอักษร

File could be viewed as array of character



```
main.pas  testLine.txt  ⋮
1  abcdefg
2  hijklmn
```

testLine.txt

Index ---->	0	1	2	3	4	5
str ---->	H	O	M	E	\0	

Address ---->	0x23451	0x23452	0x23453	0x23454	0x23455
---------------	---------	---------	---------	---------	---------

Type

Index: Integer
 c1,c2,c3:character
 str : string

str := "HOME";
 Index := 1;

c1 := str[index];
 index := index+1;

c2 = str[index];
 index := index+1;

Index := 4;
 c3 = str[index];
 index := index+1;

Read 'H'
 adjust position forward (2)

Read 'O'
 adjust position forward (3)

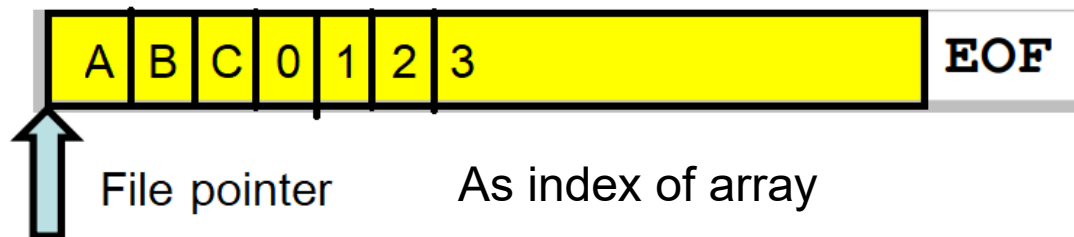
read 'E'
 adjust position backward

readtestLine.pas

```

1 program ReadFileWithPosition;
2 uses
3   SysUtils;
4 var
5   fp1: File;
6   ch: Byte;
7   pos: Int64;
8   bytesRead: Integer;
9 begin
10  Assign(fp1, 'testLine.txt');
11  Reset(fp1, 1);
12  while True do
13  begin
14    pos := FilePos(fp1);
15    BlockRead(fp1, ch, 1, bytesRead);
16    if bytesRead = 0 then
17      Break;
18    Writeln(Char(ch), ' ', IntToHex(
19      end;
20
21  Writeln;
22  Writeln('data @pos 3');
23  Seek(fp1, 3);
24  pos := FilePos(fp1);
25  BlockRead(fp1, ch, 1, bytesRead);
26  Writeln(Char(ch), ' ', IntToHex(ch
27  Close(fp1);
28 end.
```

File could be viewed as array of character



readtestLine.pas

```
1 program ReadFileWithPosition;
2 uses
3   SysUtils;
4 var
5   fp1: File;
6   ch: Byte;
7   pos: Int64;
8   bytesRead: Integer;
9 begin
10  Assign(fp1, 'testLine.txt');
11  Reset(fp1, 1);
12  while True do
13  begin
14    pos := FilePos(fp1);
15    BlockRead(fp1, ch, 1, bytesRead);
16    if bytesRead = 0 then
17      Break;
18    Writeln(Char(ch), ' ', IntToHex(ch,2), ' ', pos);
19  end;
20
21  Writeln;
22  Writeln('data @pos 3');
23  Seek(fp1, 3);
24  pos := FilePos(fp1);
25  BlockRead(fp1, ch, 1, bytesRead);
26  Writeln(Char(ch), ' ', IntToHex(ch,2), ' ', pos);
27  Close(fp1);
28 end.
```

```
Target OS: Linux fo
Compiling main.pas
Linking a.out
33 lines compiled,
0 30 0
1 31 1
2 32 2
3 33 3
4 34 4
5 35 5
6 36 6
7 37 7
8 38 8
9 39 9
0 30 10
a 61 11
b 62 12
c 63 13
d 64 14
e 65 15
f 66 16
g 67 17

data @pos 3
3 33 3
```

Array vs File concepts

■ Array of character

Type

Index: Integer
c1,c2:character
str : string

str := "HOME";
Index := 1;

c1 := str[index];
index := index+1;

c2 = str[index];
index := index+1;

read
adjust position forward

index := 3
str[index] = 'A'
index := index+1;

write
adjust position backward

```
1 program ReadFileWithPosition;
2 uses
3   SysUtils;
4 var
5   fp1: File;
6   ch: Byte;
7   pos: Int64;
8   bytesRead: Integer;
9 begin
10  Assign(fp1, 'testLine.txt');
11  Reset(fp1, 1);
12  while True do
13  begin
14    pos := FilePos(fp1);
15    BlockRead(fp1, ch, 1, bytesRead);
16    if bytesRead = 0 then
17      Break;
18    Writeln(Char(ch), ' ', IntToHex(pos, 16));
19  end;
20
21  Writeln;
22  Writeln('data @pos 3');
23  Seek(fp1, 3);
24  pos := FilePos(fp1);
25  BlockRead(fp1, ch, 1, bytesRead);
26  Writeln(Char(ch), ' ', IntToHex(pos, 16));
27  Close(fp1);
28 end.
```

ลำดับการทำงานแฟ้มข้อมูล

1. เปิดแฟ้มข้อมูล

1.1. Assign (f, 'myfile.dat')

Reset (f) /Rewrite (f) /Append(f)

2. การจัดการบนแฟ้มข้อมูล

2.1. อ่าน

2.1. BlockRead() ,Read()

2.2. เขียน

2.2. BlockWrite(), Write(), Append()

2.3. การสืบค้นตำแหน่ง

2.3. Seek() , FilePost ()

3. ปิดแฟ้มข้อมูล

3. close()

Text File Manage

คำสั่งการจัดการแฟ้มแบ่งออกเป็น 3 กลุ่ม (Based on FPC)

- Text File Mode (TEXT)
- Binary File Mode (FILE)

ลำดับ operation บน text file

0: สร้าง file pointer

1. กำหนดชื่อแฟ้ม

2. เปิดแฟ้มข้อมูล

3. การจัดการบนแฟ้มข้อมูล

3.1. อ่าน

3.2. เขียน

~~3.3. การสลับตำแหน่ง~~
~~(ทำไม่ได้)~~

4. ปิดแฟ้มข้อมูล

0. var f : Text

1. assign(f,'mydata')

2.1. Reset(f) / เปิดเพื่ออ่าน

2.1. Append(f) / เปิดเขียนแบบต่อท้าย

2.2. Rewrite(f) / เปิดเขียนแบบ overwrite

3.1 Read(f, var1, var2, ...);

ReadLn(f, var1, var2, ...);

3.2 WriteLn(f, 'New line');

Write(f, 'line');

4. Close(f);

Opening/Close a File

- A file must be “opened” before it can be used.

```
Var fi : Text
```

```
Assign( fi, 'ReadFile');  
Reset( fi );
```

```
Close( fi );
```

```
Var fi : Text
```

```
Assign( fi, 'overwriteFile');  
Rewrite( fi );
```

```
Close(fi );
```

```
Var fi : Text
```

```
Assign( fi, 'AppendFile');  
Append( fi );
```

```
Close( fi );
```

Files and Streams

- Read/Readln : ต้องมีรูปแบบ

Var

fi:text

str3: string[3+1]

i1 : integer;

f : real;

str2 : string[2+1];

i2 : integer

Readln(fi , str1 , i , f , str2 , i2);

A01 123 40.5 B1 13

A02 124 40.1 B2 14

Files and Streams

■ Read/Readln :

1. ต้องมีรูปแบบข้อมูลเหมือนกันทั้ง file

A01 123 40.5 B1 13

A02 124 40.1 B2 14

2. แต่ละ field มีขนาดคงที่ มี space ขึ้น

3. กรณี field มีชนิดเป็น string

3.1. ขนาดของ field ต้องเท่ากันทุกๆ record

3.2. ต้องจองพื้นที่ของตัวแปรมีขนาดเท่ากับ field+1

เช่น A02 -> ตัวแปรที่อ่านต้องมีขนาด `string[3+1];`

B1 -> ตัวแปรที่อ่านต้องมีขนาด `string[2+1];`

main.pas

teacherInfo.txt

```
1 program ReadTeacherInfo;
2
3 var
4   f: Text;
5   id: string[9+1]; //add 1 for spc
6   no: Integer;
7   level: string[1+1]; //add 1 for spc
8   grade: real;
9   score: Integer;
10  code: string[4+1]; //add 1 for spc
11  group: Integer;
12  //T65001001 1 A 1.1 10 A001 -3
13 begin
14   Assign(f, 'teacherInfo.txt');
15   Reset(f); { เปิดไฟล์เพื่ออ่าน }
16
17   while not Eof(f) do
18   begin
19     ReadLn(f, id, no, level, grade, score, code, group);
20
21     Writeln('ID      : ', id);
22     Writeln('No      : ', no);
23     Writeln('level   : ', level);
24     Writeln('Grade  : ', grade:0:2);
25     Writeln('Score  : ', score);
26     Writeln('Code   : ', code);
27     Writeln('group  : ', group);
28     Writeln('-----');
29   end;
30
31   Close(f); { ปิดไฟล์ }
32 end.
```

```
1 T65001001 · 1 · A · 1.1 · 10 · A001 · -3
2 T65001002 · 2 · B · 2.1 · 20 · A002 · 2
3 T65001003 · 3 · C · 3.1 · 20 · A003 · -1
4 T65001004 · 4 · D · 4.1 · 40 · A004 · 0
5 T65001005 · 5 · A · 5.1 · 50 · A005 · -1
6 T65001006 · 6 · A · 6.1 · 60 · A006 · 3
7 T65001007 · 7 · A · 7.1 · 70 · A007 · 2
8 T65001008 · 8 · B · 8.1 · 80 · A008 · 1
```

reafmt.pas

teacherInfo.txt

Write Operations on Files

Write/WriteLn

var fi:Text;

Write(fi, var1, ' ', "const-string" , ' ', 1, ' ', 3.2);

writfmt.pas

```
1 program ReadTeacherInfo;
2
3 var
4   f1,f2: Text;
5   id: string[9+1]; //add 1 for spc
6   no: Integer;
7   level: string[1+1]; //add 1 for spc
8   grade: real;
9   score: Integer;
10  code: string[4+1]; //add 1 for spc
11  group: Integer;
12  //T65001001 1 A 1.1 10 A001 -3
13 begin
14   Assign(f1, 'teacherInfo.txt');
15   Reset(f1); { เปิดไฟล์เพื่ออ่าน }
16
17   Assign(f2, 'teacherInfo.out');
18   //Rewrite(f2);
19   Append(f2);
20
21   while not Eof(f1) do
22   begin
23     ReadLn(f1, id, no, level, grade, score, code, group);
24
25     Writeln('ID      : ', id);
26     Writeln('No      : ', no);
27     Writeln('level   : ', level);
28     Writeln('Grade  : ', grade:0:2);
29     Writeln('Score  : ', score);
30     Writeln('Code   : ', code);
31     Writeln('group  : ', group);
32     Writeln('-----');
33     if group > 0 then
34     Writeln(f2, id, ' ', no, ' ', level, ' ', grade:0:2, ' ', score, ' ', code, ' ', group);
35   end;
36
37   Close(f1); { ปิดไฟล์ }
38   Close(f2);
39 end.
```

Operation on File

- Read Only
 - Create a new File or over write exist File
 - Read/Write File (update file)
 - Append File
-

```
1 program MergeTwoFiles;
2 var
3   fileA, fileB, fileC: Text;
4   num1, num2: Integer;
5   f1, f2: Boolean;
6
7 begin
8   Assign(fileA, 'class1.txt');
9   Reset(fileA);
10
11   Assign(fileB, 'class2.txt');
12   Reset(fileB);
13
14   Assign(fileC, 'class3.out');
15   Rewrite(fileC);
16
17   Read(fileA, num1);
18   Read(fileB, num2);
19
20   while not Eof(fileA) and not Eof(fileB) do
21   begin
22     if num1 < num2 then
23     begin
24       Writeln(fileC, num1);
25       Read(fileA, num1);
26     end
27     else if num2 < num1 then
28     begin
29       Writeln(fileC, num2);
30       Read(fileB, num2);
31     end
32     else
33     begin
34       Writeln(fileC, num1);
35       Read(fileA, num1);
36       Read(fileB, num2);
37     end;
38   end;
39
40   Close(fileA);
41   Close(fileB);
42   Close(fileC);
43 end.
```

1
3
5
6
7
9
11
13
12

2
4
6
8
10
12
14
16

simmerg.pas

ตัวอย่าง 1

สมมติว่าข้อมูล อาจารย์คณะวิทยาศาสตร์มี 150 ดังนี้

เพศ (sex) : ชาย = 1 , หญิง = 2

อายุ (age) : ? ปี

สถานภาพสมรส (status) : โสด = 1 , แต่งงาน = 2 , อื่นๆ = 2

ปริญญาสูง (degree) : ตรี = 1 , โท = 2 , เอก = 3

จำนวนปีที่สอน (experience) : ? ปี

จงเขียน

1. โปรแกรมรับข้อมูลของอาจารย์คณะวิทยาศาสตร์จำนวน 150 คน

2. หาจำนวนปีเฉลี่ยของ อายุ และ จำนวนปีที่สอน

AVEREAGE AGE = xx.xx Years

AVEREAGE Experience = XX.XX Years

main.pas

```

1 program EmployeeAverage;
2 const numPerson = 3; //ค่าคงที่
3 type
4   Employee = record
5     sex: Integer;
6     age: Integer;
7     stat: Integer;
8     degr: Integer;
9     tyear: Integer;
10  end;
11
12 var
13   person: array[1..numPerson] of Employee;
14   i: Integer;
15   avg_age, avg_expr: Real;
16   numAgeAbove, numYearAbove: Integer;
17
18 begin
19   { Input data for first 2 employees }
20   for i := 1 to numPerson do
21     begin
22       Write('sex(1-2) = '); ReadLn(person[i].sex);
23       Write('age = '); ReadLn(person[i].age);
24       Write('stat (1-3) = '); ReadLn(person[i].stat);
25       Write('degr(1-3) = '); ReadLn(person[i].degr);
26       Write('tyear = '); ReadLn(person[i].tyear);
27     end;
28
29   avg_age := 0;
30   avg_expr := 0;

```

```

29   avg_age := 0;
30   avg_expr := 0;
31
32   { Compute sums }
33   for i := 1 to numPerson do
34     begin
35       avg_age := avg_age + person[i].age;
36       avg_expr := avg_expr + person[i].tyear;
37     end;
38
39   avg_age := avg_age / numPerson;
40   avg_expr := avg_expr / numPerson;
41
42   numAgeAbove := 0;
43   numYearAbove := 0;
44
45   { Count how many are above average }
46   for i := 1 to numPerson do
47     begin
48       if person[i].age > avg_age then
49         numAgeAbove := numAgeAbove + 1;
50
51       if person[i].tyear > avg_expr then
52         numYearAbove := numYearAbove + 1;
53     end;
54
55   { Output results }
56   WriteLn('AVG AGE = ', avg_age:0:2);
57   WriteLn('NUMBER OF AGE ABOVE AVERAGE = ', numAgeAbove);
58   WriteLn('AVG EXPR = ', avg_expr:0:2);
59   WriteLn('NUMBER OF YEAR ABOVE AVERAGE = ', numYearAbove);
60 end.

```

ตัวอย่าง 1

สมมติว่าข้อมูล อาจารย์คณะวิทยาศาสตร์มี 150 ดังนี้

เพศ (sex) : ชาย = 1 , หญิง = 2

อายุ (age) : ? ปี

สถานภาพสมรส (status) : โสด = 1 , แต่งงาน = 2 , อื่นๆ = 2

ปริญญาสูง (degree) : ตรี = 1 , โท = 2 , เอก = 3

จำนวนปีที่สอน (experience) : ? ปี

จงเขียน

1. โปรแกรมรับข้อมูลของอาจารย์คณะวิทยาศาสตร์จำนวน 150 คน
2. หาจำนวนปีเฉลี่ยของ อายุ และ จำนวนปีที่สอน

AVERAGE AGE = XX.XX Years

AVERAGE Experience = XX.XX Years

T65001001, รหัสอาจารย์ (2)

1, เพศ

20, จำนวนปีที่สอน

1, ระดับปริญญา

1 สถานภาพสมรส

ให้นักศึกษา เขียนโปรแกรมอ่านข้อมูลอาจารย์จากแฟ้ม (file) ซึ่งมีรูปแบบ (format) การจัดเก็บดังนี้

T65001001 1 20 1 1

T65001002 2 22 2 2

T65001003 1 21 2 2

T65001004 2 22 1 2

T65001005 2 23 3 2

T65001006 2 19 3 2

empData.txt

main.pas

empData.txt

class1.txt

class2.txt

class3.out

teacherInfo.txt

```
1  program TeacherInfo;
2  type
3      employee = record
4          Tcode : string[9+1];
5          sex   : Integer;
6          tyear : Integer;
7          degr  : Integer;
8          stat  : Integer;
9      end;
10
11 var
12     person : array[1..150] of employee;
13     f       : Text;
14     i, count : Integer;
15     avg_expr : Real;
16
17 begin
18     Assign(f, 'empData.txt');
19     Reset(f);
20
21     Writeln('code':12, 'sex':6, 'year':6, 'degr':6, 'stat':6);
22
23     count := 0;
```

FileEmpAvg.pas

empData.txt


```
25 while not Eof(f) do
26 begin
27
28     count := count+1;
29
30     ReadLn(f,
31         person[count].Tcode,|
32         person[count].sex,
33         person[count].tyear,
34         person[count].degr,
35         person[count].stat);
36
37     Writeln(person[count].Tcode:12,
38         person[count].sex:6,
39         person[count].tyear:6,
40         person[count].degr:6,
41         person[count].stat:6);
42 end;
43
44 Close(f);
45
46 { คำนวณค่าเฉลี่ยปีที่สอน }
47 avg_expr := 0;
48 for i := 1 to count do
49     avg_expr := avg_expr + person[i].tyear;
50
51 if count > 0 then
52     avg_expr := avg_expr / count;
53
54 Writeln;
55 Writeln('AVG EXPR = ', avg_expr:0:2);
56
57 end.
```



END



Binary File

ลำดับ operation บน text file

0: สร้าง file pointer

1. กำหนดชื่อแฟ้ม

2. เปิดแฟ้มข้อมูล

3. การจัดการบนแฟ้มข้อมูล

3.1. อ่าน

3.2. เขียน

3.3. การสื่บค้นตำแหน่ง

4. ปิดแฟ้มข้อมูล

0. var f : Text

1. assign(f,'mydata')

2.1. Reset(f) / เปิดเพื่ออ่าน

2.1. Append(f) / เปิดเขียนแบบต่อท้าย

2.2. Rewrite(f) / เปิดเขียนแบบ overwrite

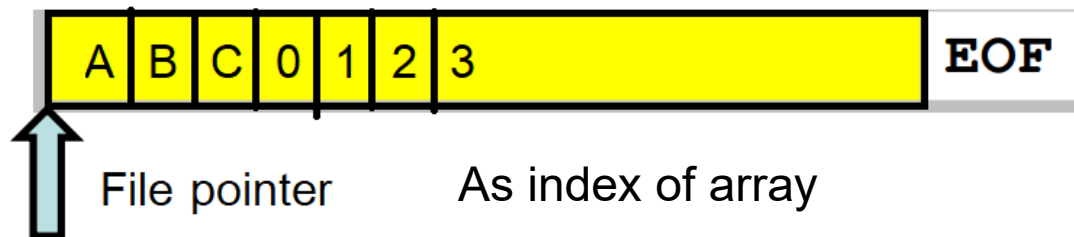
3.1 BlockRead (f, var1, ...);

3.2 BlockWrite(f, var1,...);

FilePost(f) , Seek()

4. Close(f);

File could be viewed as array of character



readtestLine.pas

```
1 program ReadFileWithPosition;
2 uses
3   SysUtils;
4 var
5   fp1: File;
6   ch: Byte;
7   pos: Int64;
8   bytesRead: Integer;
9 begin
10  Assign(fp1, 'testLine.txt');
11  Reset(fp1, 1);
12  while True do
13  begin
14    pos := FilePos(fp1);
15    BlockRead(fp1, ch, 1, bytesRead);
16    if bytesRead = 0 then
17      Break;
18    Writeln(Char(ch), ' ', IntToHex(ch,2), ' ', pos);
19  end;
20
21  Writeln;
22  Writeln('data @pos 3');
23  Seek(fp1, 3);
24  pos := FilePos(fp1);
25  BlockRead(fp1, ch, 1, bytesRead);
26  Writeln(Char(ch), ' ', IntToHex(ch,2), ' ', pos);
27  Close(fp1);
28 end.
```

```
Target OS: Linux fo
Compiling main.pas
Linking a.out
33 lines compiled,
0 30 0
1 31 1
2 32 2
3 33 3
4 34 4
5 35 5
6 36 6
7 37 7
8 38 8
9 39 9
0 30 10
a 61 11
b 62 12
c 63 13
d 64 14
e 65 15
f 66 16
g 67 17

data @pos 3
3 33 3
```

l	6c	0	
i	69	1	
n	6e	2	
e	65	3	rb
-	2d	4	
1	31	5	
	d	6	

	a	7	
l	6c	8	
i	69	9	
n	6e	10	
e	65	11	
-	2d	12	
2	32	13	
	d	14	
	a	15	
l	6c	16	
i	69	17	

l	6c	0	
i	69	1	
n	6e	2	
e	65	3	rt
-	2d	4	
1	31	5	

	a	6	
l	6c	8	
i	69	9	
n	6e	10	
e	65	11	
-	2d	12	
2	32	13	

	a	14	
l	6c	16	
i	69	17	
n	6e	18	
e	65	19	

File	Edit	Options	Encoding	Help														
00000000:	6C	69	6E	65	2D	31	0D	0A	6C	69	6E	65	2D	32	0D	0A		line-1..line-2..
00000010:	6C	69	6E	65	2D	33	0D	0A	6C	69	6E	65	2D	34	0D	0A		line-3..line-4..

	binary	23	
l	6c	24	
i	69	25	
n	6e	26	

l	text	25	
n	6e	26	
e	65	27	
-	2d	28	
4	34	29	

RUN AN...  No Config  ...

VARIABLES

Locals

pos = 14

ch = 10 '\n'

> fp1 = 0x75acc0

pos #2 = 14

> Registers

WATCH

ch = 10 '\n'

CALL STACK

[1] PAUSED ON BREAKPOINT

main() file.c 14:1

> [2] PAUSED

C file.c 5 X

C file.c > main()

2 int main()

6 if (fp1 == NULL) {
7 printf("cannot open file !!");

8 return 0;

9 }

10 do {

11 long pos = ftell(fp1);

12 ch = getc(fp1);

13 printf("%c \t %x \t %d \n",ch, ch , pos);

14 } while (ch!=EOF);

15 fclose(fp1);

16

17 }

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\mydoc\cos2101\apirak2\lab\lab9.file> & 'c:\Users\Apirak\.vscode\extension\ncher.exe' '--stdin=Microsoft-MIEngine-In-sufcrlfi.14m' '--stdout=Microsoft-MIE
'--pid=Microsoft-MIEngine-Pid-svjnozm.gck' '--dbgExe=c:\msys64\ucrt64\bin\gdb

l	6c	0
i	69	1
n	6e	2
e	65	3
-	2d	4
1	31	5

	a	6
l	6c	8

ฟังก์ชัน fwrite

int fwrite (void **buffer*, int *size*, int *count*, FILE **stream*);

input

buffer Storage location for data
size Item size in bytes (sizeof data type)
count Maximum number of items to be written
stream Pointer to **FILE** structure

return

The **fwrite** function writes up to *count* items, of *size* length each, from *buffer* to the output *stream*.

```
118 struct.mystruct.{
119     —>int.i;
120     —>char.ch;
121     —>char.str[3];
122 };
123
124 int.main(void)
125 {
126     —>FILE.*stream;
127     —>struct.mystruct.s[10].={.0.};
128     —>if.((stream.=.fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/TEST.$$$", "wb")).==.NULL).{
129     —>—>fprintf(stderr, "Cannot open output file.\n");
130     —>—>return.1;
131     —>}
132     —>for.(int.i.=.0;.i.<.10;.i++).{
133     —>—>s[i].i.=.i;
134     —>—>s[i].ch.=.'A'+.i;
135     —>—>strcpy(s[i].str, "12");
136     —>—>printf("\n%d, %c, %s", s[i].i, s[i].ch, s->str);
137     —>}
138     —>for.(int.i.=.0;.i.<.5;.i++).{
139     —>—>fwrite(&s[i], sizeof(struct.mystruct), 1, stream);
140     —>}
141
142     —>/* write struct.s to file */
143     —>fwrite(&s[5], sizeof(struct.mystruct), 5, stream);
144     —>fclose(stream); /* close file */
145     —>return.0;
146
147 }
```

```

4 struct mystruct {
5     int i;
6     char ch;
7     char str[3];
8 };
9

```

File Edit Options Encoding Help

```

00000000: 00 00 00 00 41 31 32 00|01 00 00 00 42 31 32 00 | |....A12.....B12.
00000010: 02 00 00 00 43 31 32 00|03 00 00 00 44 31 32 00 | | ...C12.....D12.
00000020: 04 00 00 00 45 31 32 00|05 00 00 00 46 31 32 00 | | ....E12.....F12.
00000030: 06 00 00 00 47 31 32 00|07 00 00 00 48 31 32 00 | | ....G12.....H12.
00000040: 08 00 00 00 49 31 32 00|09 00 00 00 4A 31 32 00 | | ....I12.....J12.

```

ฟังก์ชัน fread

```
int fread ( void *buffer, int size, int count, FILE *stream );
```

input

buffer Storage location for data
size Item size in bytes (sizeof data type)
count Maximum number of items to be read
stream Pointer to **FILE** structure

return

The **fread** function reads up to *count* items of *size* bytes from the input *stream* and stores them in *buffer*.

fread() and fwrite()

```
size_t fread(void *buffer, size_t numbytes, size_t count, FILE *a_file);
```

```
size_t fwrite(void *buffer, size_t numbytes, size_t count, FILE  
*a_file);
```

- ❑ Buffer in fread is a pointer to a region of memory that will receive the data from the file. Buffer in fwrite() is a pointer to the information that will be written to the file.
- ❑ The second argument is the size of the element; it is in bytes. For example, if you have an array of characters, you would want to read it in one byte chunks, so `numbytes` is one. You can use the `sizeof` operator to get the size of the various datatypes; for example, if you have a variable, `int x`; you can get the size of `x` with `sizeof(x)`;

Contd..

- ❑ The third argument is simply how many elements you want to read or write; for example, if you pass a 100 element array
- ❑ The final argument is simply the file pointer
- ❑ `Size_t` is an unsigned integer.
- ❑ `fread()` returns number of items read and `fwrite()` returns number of items written
- ❑ To check to ensure the end of file was reached, use the `feof` function, which accepts a `FILE` pointer and returns true if the end of the file has been reached.

```

193 struct mystruct {
194     → int i;
195     → char ch;
196     → char str[3];
197 };
198
199 int main(void)
200 {
201     → FILE *stream;
202     → //read/write where the file pointer location is at begining of file
203     → if ((stream = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/TEST. $$$", "a+b")) == NULL) {
204         → fprintf(stderr, "Cannot open output file.\n");
205         → return 1;
206     }
207
208     → while (true)
209     {
210         → struct mystruct itm;
211         → int recordno = 0;
212
213         → cout << "\nfinding record no. (0=terminate) (1-9) :: ";
214         → cin >> recordno;
215         → recordno -= 1;
216         → if (recordno < 0)
217             → break;
218
219         → fseek(stream, recordno * sizeof(struct mystruct), SEEK_SET);
220         → fread(&itm, sizeof(struct mystruct), 1, stream);
221
222         → printf("\n%d, %c, %s -> record no :: ", itm.i, itm.ch, itm.str, recordno);
223
224     }
225     → fclose(stream); /* close file */
226
227     → return 0;
228 }

```

```
1  #define _CRT_SECURE_NO_WARNINGS
2  #include <iostream>
3  #include <stdio.h>
4  #include <conio.h>
5
6  struct prod {
7      int cat_num;
8      float cost;
9  };
10
11 int main()
12 {
13     FILE *fp;
14     struct prod product;
15
16     struct prod a[3] = { {2,20.1},{4,40.1},{6,60.1} };
17     struct prod k, *p = &k;
18
19     fp = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/fread1.txt", "w+");
20     // write the entire array into the file pointed to by fp
21     fwrite(a, sizeof(product), 3, fp);
22     // prepare for reading from the beginning of the file
23     fseek(fp, SEEK_SET, 0);
24     // read from the file one product at a time
25     for (int i = 0; i < 3; i++) {
26         fread(p, sizeof(product), 1, fp);
27         printf(" product %d, cat_num=%d, cost=%f\n", i, p->cat_num, p->cost);
28     }
29     fclose(fp);
30 }
```

Rewrite the code saving data file

ตัวอย่าง

ให้นักศึกษาเขียนโปรแกรมขายสินค้าและพิมพ์ใบเสร็จรับเงิน
โดยข้อมูลในระบบมีไม่เกิน **100** รายการ

ข้อมูลขายสินค้าด้วย

- รหัสสินค้า(id) เช่น “AB001”
- ชื่อสินค้า(name) เช่น “CAFÉ”
- ราคาขาย(cost) เช่น 30.6
- จำนวนสินค้าในสต็อก (instock) เช่น 5

Menu

1. Add item to Cart
 2. Sell Product
 3. Check Stock
 4. End Program
- Please select 1 / 2 / 3/ 4 =?

กด 1

Input your product data:

id : **AB001**

name : **CAFE**

Cost : **30.6**

Stock : **5**

กด 2

Input your product id: **AB001**

Input your sell volume : **2**

name : **CAFE**

Cost : **30.6**

stock : **3**

Pay : 61.2 bath

Menu

1. Add item to Cart
2. Sell Product
3. Check Stock
4. End Program

Please select 1 / 2 / 3/ 4 =?

กด 3

product List:

Id	name	cost	stock
----	------	------	-------

AB001	CAFÉ	30.6	3
-------	------	------	---

กด 4

exit from program

```
4
5 using namespace std;
6
7 struct product {
8     char id[10];
9     char name[30];
10    float cost;
11    int instock;
12 };
13
14 int main()
15 {
16    struct product item[100];
17    int n = 0;
18    int select = -1;
19
20    while (true)
21    {
22
23        cout << " Menu" << endl;
24        cout << "1. Add item to Cart " << endl;
25        cout << "2. Sell product " << endl;
26        cout << "3. Check stock " << endl;
27        cout << "4. End program" << endl;
28        cout << "Please select 1 / 2 / 3/ 4 =? ";
29        cin >> select;
30
31        if (select == 4) {
32            break;
33        }
34        switch (select)
35        {
36            case 1:
```

single module Version

แก้ไขให้เป็น dynamic

Exercises

1. Write a C language program to read “**mark.dat**” file containing rollno, name, marks of three subjects and calculate total mark, result in grade and store same in “**result.dat**” file. (Note : Make use of fread and fwrite functions)
2. Write a C language program to read a **cust.dat** file containing meter number, name, current reading & previous reading. Read the same file. Calculate unit and total amounts according to the following rules —

Unit	rate
0-50	1.00
51-100	1.50
> 100	2.00

Store meter number ,name ,unit & amount in **master.dat** file.

END

Lister - [d:\mydoc\cos210

File Edit Options Encodi

line-1

line-2

line-3

line-4

Lister - [d:\mydoc\cos2101\apirak2\lab\lab9.file\test1.txt]

File Edit Options Encoding Help

00000000: 6C 69 6E 65 2D 31 0D 0A | 6C 69 6E 65 2D 32 0D 0A | line-1..line-2..

00000010: 6C 69 6E 65 2D 33 0D 0A | 6C 69 6E 65 2D 34 0D 0A | line-3..line-4..

```
00000000: 6C 69 6E 65 2D 31 0D 0A | line-1..line-2..
00000010: 6C 69 6E 65 2D 33 0D 0A | line-3..line-4..
```

```
1 6c 0
i 69 1
n 6e 2
e 65 3
_ 2d 4
1 31 5

1 a 6
i 6c 8
n 69 9
e 6e 10
_ 65 11
2 2d 12
3 32 13

1 a 14
i 6c 16
n 69 17
e 6e 18
_ 65 19
3 2d 20
4 33 21

1 a 22
i 6c 24
n 69 25
e 6e 26
_ 65 27
4 2d 28
34 29
```

```
7 int main()
8 {
9     char ch;
10    FILE *fp1 = fopen( "D:/mydoc/cos2101/DemoCOS2101/Debug/test1.txt", "rt");
11    if (fp1 == NULL) {
12        printf("cannot open file !!");
13        return 0;
14    }
15    do {
16        long pos = ftell(fp1);
17        ch = getc(fp1);
18        printf("%c \t %x \t %d \n",ch, ch , pos);
19    } while (ch != EOF);
20
21    fclose(fp1);
22
23 }
```

```
line-1
line-2
line-3
line-4
```

<https://www.jdoodle.com/online-compiler-c++14>

ฟังก์ชัน ftell

long ftell(FILE **stream*);

input

stream Pointer to **FILE** structure

return

returns the current file position. The value returned by **ftell** may not reflect the physical byte offset for streams opened in text mode

ฟังก์ชัน fseek

```
int fseek( FILE *stream, long offset, int origin );
```

input

stream Pointer to **FILE** structure

offset Number of bytes from *origin*

origin Initial position

- **SEEK_CUR** Current position of file pointer
- **SEEK_END** End of file
- **SEEK_SET** Beginning of file

return

0 if successful,

ตัวอย่าง

FILE *fp ;

fp = fseek(fp , 0 , SEEK_SET); */* go to top of file */*

fp = fseek(fp , 5 , SEEK_SET); */* go to 5 byte from top file */*

fp = fseek(fp , 0 , SEEK_END); */* go to button of file */*

fp = fseek(fp , - 5 , SEEK_END); */* go back 5 byte from button */*

fp = fseek(fp , 5 , SEEK_CUR); */* skip next 5 byte from cur pos */*

fp = fseek(fp , -5 , SEEK_CUR); */* go back 5 byte from curs pos*/*

fseek

```
5 int main()
6 {
7     char ch;
8     FILE *fp1 = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/test1.txt", "rt");
9     if (fp1 == NULL) {
10         printf("cannot open file !!");
11         return 0;
12     }
13
14     do {
15         long pos1;
16         printf("seek position (-1 exit) : ");
17         scanf("%d", &pos1);
18         if (pos1 < 0) {
19             break;
20         }
21         fseek(fp1, pos1, SEEK_SET);
22         long pos2 = ftell(fp1);
23         ch = getc(fp1);
24         printf("%c \t %x \t %d \n", ch, ch, pos2);
25     } while (true);
26
27     fclose(fp1);
28 }
29
30
```

NOTE: เพิ่มแบบ text mode

ลักษณะของเพิ่มแบบ text โหมด

1. มี CTRL-Z บอกเครื่องหมายจบเพิ่ม
2. เครื่อง carriage return–linefeed (0xa,0xd) ถูกแทนด้วย linefeed (0xd)
3. การใช้คำสั่งค้นหาตำแหน่งข้อมูล (fseek,ftell) อาจคลาดเคลื่อน
4. ถ้าใช้ฟังก์ชัน `fopen( const char *filename, const char *mode )` ค่าของ `mode` ต้องมี option “ t ” หรือไม่ option “ b ”

ตัวอย่าง

```
FILE *fp ;
```

```
fp = fopen( “c:\myfile.txt” , “rt” );    → Text File
```

```
fp = fopen( “c:\myfile.txt” , “rb” );    → Binary File
```

Basic File Steps to Handle File

- Opening a file
- Reading data from a file
- Writing data to a file
- Closing a file

```
7  int main()
8  {
9      char ch;
10     FILE *fp1 = fopen( "D:/mydoc/cos2101/DemoCOS2101/Debug/test1.txt", "rt");
11     if (fp1 == NULL) {
12         printf("cannot open file !!");
13         return 0;
14     }
15     do {
16         long pos = ftell(fp1);
17         ch = getc(fp1);
18         printf("%c \t  %x \t %d \n",ch, ch , pos);
19     } while (ch != EOF);
20
21     fclose(fp1);
22
23 }
```



Example: Merge two files

```
#include <stdio.h>
int main()
{
    FILE *fileA, /* first input file */
          *fileB, /* second input file */
          *fileC; /* output file to be created */
    int num1, /* number to be read from first file */
        num2; /* number to be read from second file */
    int f1, f2;

    /* Open files for processing */
    fileA = fopen("class1.txt", "r");
    fileB = fopen("class2.txt", "r");
    fileC = fopen("class.txt", "w");
```

```
/* As long as there are numbers in both files, read and compare
numbers one by one. Write the smaller number to the output file
and read the next number in the file from which the smaller
number is read. */
```

```
f1 = fscanf(fileA, "%d", &num1);
```

```
f2 = fscanf(fileB, "%d", &num2);
```

```
while ((f1!=EOF) && (f2!=EOF)) {
```

```
    if (num1 < num2) {
```

```
        fprintf(fileC, "%d\n", num1);
```

```
        f1 = fscanf(fileA, "%d", &num1);
```

```
    }
```

```
    else if (num2 < num1) {
```

```
        fprintf(fileC, "%d\n", num2);
```

```
        f2 = fscanf(fileB, "%d", &num2);
```

```
    }
```

```
    else { /* numbs are equal: read from both files */
```

```
        fprintf(fileC, "%d\n", num1);
```

```
        f1 = fscanf(fileA, "%d", &num1);
```

```
        f2 = fscanf(fileB, "%d", &num2);
```

```
    }
```

```
}
```



```
while (f1!=EOF){/* if reached end of second file, read
    the remaining numbers from first file and write to
    output file */
    fprintf(fileC,"%d\n", num1);
    f1 = fscanf(fileA, "%d", &num1);
}
while (f2!=EOF){ if reached the end of first file, read
    the remaining numbers from second file and write
    to output file */
    fprintf(fileC,"%d\n", num2);
    f2 = fscanf(fileB, "%d", &num2);
}

/* close files */
fclose(fileA);
fclose(fileB);
fclose(fileC);
return 0;
} /* end of main */
```